Course 502

Time 50 minutes

Marks 20

1. Briefly describe the schools of Macroeconomics chronologically. What are the causes and effects of fiscal and monetary policy according to Classical, Keynesian and neo- Keynesian synthesis? [3+2]
2. Briefly explain classical dichotomy and neutrality of money using the pillars of classical macroeconomics. [4]
3. Consider a Solow economy that is on its balanced growth path and production function follows Cobb-Douglas production function $Y(t)=F(K(t),A(t)L(t))$ .The fraction of output devoted to investment ,s, is exogenous and capital depreciation rate is δ. The growth rate of L and A are respectively n and g.
   - a) Show the impacts of changes in the saving ratio on per capita consumption. Does any steady state solution automatically imply golden rule of capital accumulation? Discuss.
   - b) Explain the long run effect of a rise in saving on output. In the vicinity of the balanced growth path, how rapidly does the economy converge to the balanced growth path?
   - c) Find the elasticity of output per unit of effective labor on the balanced growth path, y^*, with respect to the rate of population growth, n. If $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise y^*? [3+4+4]

$\lambda =$

Couse no. 502; Tutorial 02
Marks 20 ; Time 1 hour
Answer Any One

1. Consider a Ramsey-Cass-Koopmans model with the production function $Y = F(K, AL)$ and real interest rate r ; household's utility function takes the form

$$U=\int_{t=0}^{\infty} e^{-\rho t}u(c(t))\frac{L(t)}{H}dt$$

and the instantaneous utility function takes the CRRA (constant –relative –risk-aversion) utility form ; $U(c(t))= \frac{C(t)^{1-\theta}}{1-\theta}$;Where θ determine the household's willingness to shift consumption between different periods.

   a) Explain the features of the instantaneous utility function.
   b) Construct household budget constraint and show that there is no-Ponzi-game condition.
   c) Define Euler equation. Find out the Euler equation for households' maximization problem and interpret it.
   d) Describe how each of the following affects the $\dot{c} = 0$ and $\dot{k} = 0$ curves on the saddle path, and thus how they affect the balanced-growth-path values of c and k:
      (i) A rise in θ.
      (ii) A change in the rate of depreciation from the value of zero assumed in the text to some positive level.
   e) What is the difference between the balanced growth paths of the Solow and Ramsey–Cass–Koopmans?
   f) Briefly explain the effects of permanent and temporary changes in government purchases.

[2+3+4+4+2+5]

2. Suppose $Y_t = F(K_t, A_tL_t)$, with having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+ g)A_t$, $L_{t+1} = (1+n)L_t$ and $k_{t+1} = L_t(w_tA_t - C_{1t})$. Assume time is discrete rather than continuous and each individual lives for only two periods where θ determines how much individuals varies in response to differences between r (real rate of return) and ρ (discount rate).

   a) Solve the individual's maximization problem. Explain the role of income effect and substitution effect to determine individuals' consumption and saving.
   b) Find an expression for k_{t+1} as a function of k_t.
   c) Describe how each of the following affects k_{t+1} as a function of k_t:
      (i) A fall in n.
      (ii) A rise in ρ.
   d) Find the speed of convergence of k to k*. How does it differ from Solow model?
   e) What are the possibilities after relaxing the assumptions of logarithmic utility and Cobb–Douglas production?
   f) According to Abel, Mankiw, Summers, and Zeckhauser (1989), are modern dynamically efficient? Briefly explain.

[6+2+4+4+2+2]

G8

Course 502

Time 1 hour

Marks 20

Answer any one

1. Consider a simplified version of the models of R&D and growth model developed by P.Romer (1990), Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of R&D and *B* is a shift parameter. Both goods producing sector and R&D producing sector use the full stock of knowledge A. [6+4+4+6]

   a) Critically explain the role of aL (the labor force used in the R&D sector) in long run for the dynamics of knowledge accumulation without considering capital into the model .
   b) Assume that θ + β < 1 and n > 0, and that the economy is on its balanced growth path. Describe how an decrease in "n" affects the $\dot{g}A = 0$ and $\dot{g}K = 0$ lines and the position of the economy in (gA ,gK) space at the moment of the change.
   c) Distinguish between semi-endogenous growth model and fully endogenous growth model.
   d) What is the nature of knowledge and what are the determinants of the allocation of resources to R & D.

2. Suppose firm follows either production function to produce output and L(*i*) is used to denote both the quantity of labor devoted to producing input *i* and the quantity of input *i* that goes into final-goods production . Individual's discount rate is ρ and their instantaneous utility function is logarithmic. The utility of the representative household takes the constant-relative-risk aversion (CRRA) form is $\frac{\dot{C}(t)}{C(t)} = \frac{r(t)-\rho}{\theta}$ and θ is the coefficient of the relative risk aversion. Knowledge creation, $\frac{\dot{A}(t)}{A(t)} = BL_A$ .Growth rate of the wage equals the growth rate of output. Population growth is zero. [6+6+4+4]

   a) Briefly discuss the assumptions to construct the model.
   b) How the long-run growth will be determined by using the present value of profits from the discovery of a new idea at time t? What will be the equilibrium value of LA.
   c) What will be the socially optimal level of LA ? Explain how it is different from the equilibrium level of LA.
   d) Briefly explain the empirical test of endogenous growth models.

Course no 502

Tutorial 03

Marks 20 [4+6+6+4]

Time 50 minutes

The representative producer of good *i* has production function $Q_i=L_i$; utility depends on consumption and Labor effort where P is known as perfect information. z_i (good specific shock) and m (monetary shock) are normally distributed. Conditional expectation is a linear equation. On the other hand,For representative consumer $\dot{C}_t = \sigma\,(r_t - p)\,C_t$ and wicksellian interest rate

$r^n \equiv p + \sigma^{-1}g$. According to Yun (1996) Optimal price v_t is

$$v_t = p_t + \alpha \int_t^{\infty} [(v_s - p_s) + \eta x_s\,] e^{-(\alpha+p)(s-t)} ds$$

and $\kappa \equiv \alpha^2\eta$;Where η is sensitivity parameter.

a) Explain the role of rational expectation in determining the effectiveness of macro policy when there is imperfect information by using Lucas' supply curve .
b) Determine new Keynesian IS equation for continuous time.
c) Determine new Keynesian PC curve for continuous time using Leibniz's rule. Explain "divine coincidence" according to Blanchard and Gali.
d) Describe taylor rule in the canonical New Keynesian model where x (output) and π (inflation) both are jump variables.